



MANIPAL SCHOOL OF INFORMATION SCIENCES

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Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems, Dual Degree Program with ESIGELEC, France) Course Plan

Course Name	:	Embedded Systems
Course Code	:	ESE 5105
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Dr. Raghudathesh G P
Name of the Program Coordinator	:	Dr. B Dinesh Rao

	
Date:	10/7/2026
Signature of Program Coordinator	
with Date	

	
Date:	7/7/2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems, Dual Degree Program with ESIGELEC, France), program are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in embedded systems.
PEO 2	Introduce state of art technologies in the area of embedded systems and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering - M.E in Computer Science and Engineering (Embedded Systems, Dual Degree Program with ESIGELEC, France), graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop and implement embedded systems requirements based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analyzing and decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Embedded Systems [ESE 5105]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Basic of micro-controller, C Programming
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No.ofContact Hours	Program Outcomes (POs)	BL
CO1	Apply the knowledge regarding microcontrollers to build embedded systems.	9	PO3	3
CO2	Implement programs for interfacing on ARM microcontrollers.	15	PO3	3
CO3	Design multi-tasking applications using real-time operating systems.	12	PO5	6



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1.3 Assessment Plan

Components	Mid semester exam (MSE)	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	1 Month	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply	Implement, Design	Apply, Implement, design
Pattern	Answer all 5 questions of 10 marks each.	Programming on Embedded C	Answer all 10 full questions of 10 marks each.
Schedule	As per academic calendar.	Assignment submission: Novembder 2026	As per academic calendar.
Topics covered	Introduction to embedded systems, Arm processors, architectures, programing , interrupts, memory, registers, GPIO, cmsis, data acquivision system, AC, DAC, Serical communication system	Programming on Embedded C, RTOS.	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



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1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Introduction to Embedded Systems: Design Challenges, Processors Technology, Design Technology	C01
L2	Introduction to ARM Cortex Processor, Variants of Cortex and ARM versions, Comparison of M-series processors	C01
L3	Architecture, Programmer's Model, APSR Register	C01
L4	Memory Model, Exception, Interrupts, Reset	C01
L5	Exceptions and Interrupts, NVIC, Memory Protection Unit	C01
L6	Understanding CMSIS, Understanding CMSIS Architecture, Startup Code	C01
L7	Embedded C Programming	C01
L8	Embedded C Programming	C01
L9	Embedded C Programming	C01
L10	Introduction to LPC13/17xx Microcontroller: Memory Mapping, Registers involved, and programming with GPIO, PWM	C02
L11	Programming with GPIO, PWM	C02



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L12	Programming with GPIO, PWM	C02
L13	Data Acquisition System: ADC, Types of ADC, Choosing the ADC, DAC	C02
L14	Data Acquisition System: ADC, Types of ADC, Choosing the ADC, DAC	C02
L15	Data Acquisition System: ADC, Types of ADC, Choosing the ADC, DAC	C02
L16	Serial Communication: UART	C02
L17	Serial Communication: UART	C02
L18	Serial Communication: UART	C02
MSE	Mid Semester Exam	C01, C02
L19	Serial Communication: I2C, SPI, Interfacing	C02
L20	Serial Communication: I2C, SPI, Interfacing	C02
L21	Serial Communication: I2C, SPI, Interfacing	C02
L22	CAN Bus: Introduction, Frames, Bit Stuffing, Types of Errors, Nominal Bit Timing, A Simple Application with CAN	C02
L23	CAN Bus: Introduction, Frames, Bit Stuffing, Types of Errors, Nominal Bit Timing, A Simple Application with CAN	C02
L24	CAN Bus: Introduction, Frames, Bit Stuffing, Types of Errors, Nominal Bit Timing, A Simple Application with CAN	C02



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L25	USB Bus: Speed Identification on the Bus, States, Packets, Data Flow Types, Enumeration, Descriptors	C03
L26	USB Bus: USB Interface – C Programs	C03
L27	USB Bus: USB Interface – C Programs	C03
L28	Introduction to Multitasking in Microcontrollers: Variants of RTOS	C03
L29	FreeRTOS, UCOS, uClinux, FreeRTOS on Cortex-based Microcontrollers	C03
L30	FreeRTOS, UCOS, uClinux, FreeRTOS on Cortex-based Microcontrollers	C03
L31	FreeRTOS, UCOS, uClinux, FreeRTOS on Cortex-based Microcontrollers	C03
L32	Task Creation, Queues, Semaphore, Mutex, Application Development	C03
L33	Task Creation, Queues, Semaphore, Mutex, Application Development	C03
L34	Task Creation, Queues, Semaphore, Mutex, Application Development	C03
L35	Task Creation, Queues, Semaphore, Mutex, Application Development	C03
L36	Task Creation, Queues, Semaphore, Mutex, Application Development	C03



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1.5 References

1. Joseph Yiu, "The definitive guide to the ARM Cortex-M3", Elsevier, 2nd Edition, 2010.
2. Frank Vahid, Tony Givargis, "Embedded System Design: A Unified Hardware/Software Introduction", Wiley India, ISBN:81-265-0837-X, 2007.
3. Richard Barry, "NXP Semiconductors, LPC13xx/17xx User Manual", 2012.
4. NXP Semiconductors, "LPCzone Examples", 2012.
5. "FreeRTOS Reference Manual", Real Time Engineers Ltd., 2016.
6. MOOC: <https://in.coursera.org/learn/introduction-embedded-systems#syllabus>.

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



1.7 Course Timetable

I Semester						Room: LG01-LH09			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday									
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									



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1.8 Assessment Plan

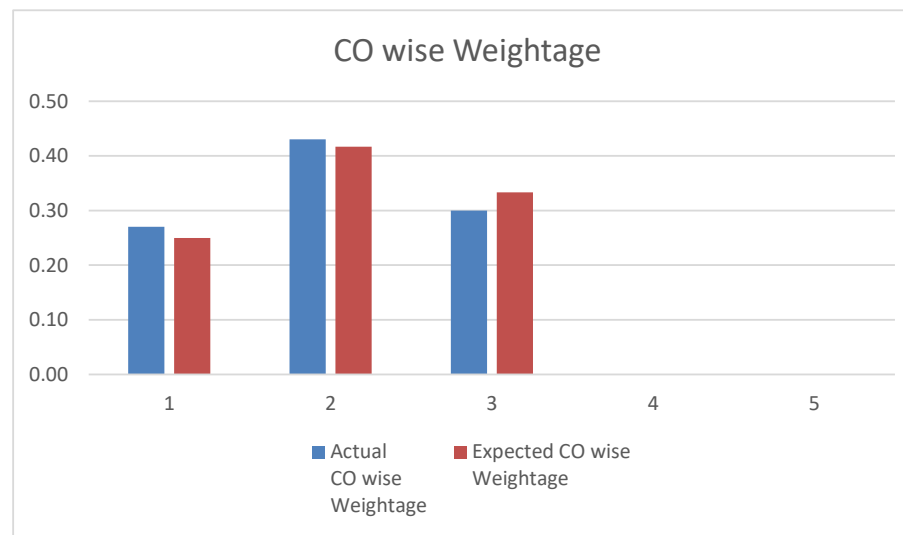
COs		Marks & Weightage				
CO No.	CO Name	MSE Marks (Max. 50)	Assignment Marks (Max. 20)	ESE Marks (Max. 100)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Apply the knowledge regarding microcontrollers to build embedded systems.	20		30	0.27	0.25
C02	Implement programs for interfacing on ARM microcontrollers.	30	10	30	0.43	0.42
C03	Design multi-tasking applications using real-time operating systems.		10	40	0.30	0.33
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes the performances in class participation, assignment work, class tests, MSE, quizzes etc.
- ESE for this course is conducted for a maximum of 100 and the same will be scaled down to 50 for grading.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

$$\text{Weightage for CO1} = (\text{MSE marks for CO1} / 1.6666 + \text{Assignment marks for CO1} / 1.0 + \text{ESE marks for CO1} / 2) / 100$$



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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	MSE	0.3	1	- Performance is measured using MSE attainment level. - Reference: question paper and answer scheme. - MSE is assessed for a maximum of 50 marks and scaled down to 30 marks.
2	Assignments	0.2	2	- Performance is measured using assignments/quiz attainment level. - Assignments/quiz are evaluated for a maximum of 20 marks.
3	ESE	0.5	1	- Performance is measured using ESE attainment level. - Reference: question paper and answer scheme. - ESE is assessed for a maximum of 100 marks and scaled down to 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2			Y		
CO3					Y
Average Articulation Level			*		*